

2201NSC Linear Algebra and Applications

Week 2

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Matrices

Dimensions, products and blocks

The size of a matrix is rows by columns

A matrix with m rows and n columns has size $m \times n$:

$$A = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix} \in \mathbb{R}^{m \times n}.$$

Example

$$\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix} \quad \text{has 2 rows and 3 columns, so its size is } 2 \times 3.$$

Addition and subtraction are entry by entry

Matrices can be added or subtracted only when they have the **same size**.

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \pm \begin{pmatrix} e & f \\ g & h \end{pmatrix} = \begin{pmatrix} a \pm e & b \pm f \\ c \pm g & d \pm h \end{pmatrix}.$$

Example

$$\begin{pmatrix} 3 & 1 \\ -2 & 4 \end{pmatrix} - \begin{pmatrix} 1 & 5 \\ 0 & 2 \end{pmatrix} = \begin{pmatrix} 2 & -4 \\ -2 & 2 \end{pmatrix}.$$

Dimensions decide whether a product exists

If $A \in \mathbb{R}^{m \times n}$ and $B \in \mathbb{R}^{p \times q}$, then

AB exists if and only if $n = p$.

Inner dimensions match; outer dimensions remain

$$\underbrace{(m \times \textcolor{brown}{n})}_A \underbrace{(\textcolor{brown}{n} \times q)}_B \longrightarrow \underbrace{(m \times q)}_{AB}.$$

The existence of AB says nothing by itself about whether BA exists.

Matrix multiplication is row by column

For $C = AB$, the entry in row i , column j is

$$c_{ij} = \sum_{k=1}^n a_{ik} b_{kj}.$$

One entry

Row i of A meets column j of B .

Whole product

Every row of A meets every column of B .

Order matters in matrix multiplication

In general,

$$AB \neq BA.$$

- ▶ One product may exist while the other does not.
- ▶ If both exist, they may have different dimensions.
- ▶ Even when both have the same size, their entries usually differ.

The identity matrix

For $A \in \mathbb{R}^{m \times n}$,

$$\boxed{I_m A = A} \quad \text{and} \quad \boxed{A I_n = A}.$$

$$I_n = \begin{pmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{pmatrix}.$$

Transpose of a matrix

If $A \in \mathbb{R}^{m \times n}$, then $A^T \in \mathbb{R}^{n \times m}$.

Transpose of a product

$$(AB)^T = B^T A^T$$

Double transpose

$$(A^T)^T = A$$

A 4×2 matrix becomes 2×4

$$\begin{pmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \\ 7 & 8 \end{pmatrix}^T = \begin{pmatrix} 1 & 3 & 5 & 7 \\ 2 & 4 & 6 & 8 \end{pmatrix}.$$

Block multiplication uses the same rule

$$\begin{pmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{pmatrix} \begin{pmatrix} B_{11} & B_{12} \\ B_{21} & B_{22} \end{pmatrix} = \begin{pmatrix} A_{11}B_{11} + A_{12}B_{21} & A_{11}B_{12} + A_{12}B_{22} \\ A_{21}B_{11} + A_{22}B_{21} & A_{21}B_{12} + A_{22}B_{22} \end{pmatrix}.$$

Why use blocks?

Identity and zero blocks can eliminate most of the work and reveal the structure of the result.

Identity and zero blocks expose the answer

$$A = \begin{pmatrix} 1 & 0 & 0 & 2 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} I_3 & P \\ 0 & I_1 \end{pmatrix}, \quad P = \begin{pmatrix} 2 \\ 0 \\ 3 \end{pmatrix}. \quad B = \begin{pmatrix} 1 & -1 \\ 2 & 0 \\ 0 & 4 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} Q \\ 0 \end{pmatrix}, \quad Q = \begin{pmatrix} 1 & -1 \\ 2 & 0 \\ 0 & 4 \end{pmatrix}.$$

$$\begin{pmatrix} I_3 & P \\ 0 & I_1 \end{pmatrix} \begin{pmatrix} Q \\ 0 \end{pmatrix} = \begin{pmatrix} I_3 Q + P 0 \\ 0 Q + I_1 0 \end{pmatrix} = \begin{pmatrix} Q \\ 0 \end{pmatrix}.$$

Short Concept Quiz

No calculations required

Question 1

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix}, \quad B = \begin{pmatrix} 0 & 1 & 0 \\ 2 & 0 & 2 \end{pmatrix}, \quad C = \begin{pmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{pmatrix}.$$

Which matrices can be added together?

- A. A and B only
- B. A and C only
- C. B and C only
- D. All three matrices

Question 2

$$A \in \mathbb{R}^{3 \times 5}, \quad B \in \mathbb{R}^{5 \times 2}, \quad C = AB.$$

What are the dimensions of C ?

- A. $C \in \mathbb{R}^{3 \times 2}$
- B. $C \in \mathbb{R}^{5 \times 5}$
- C. $C \in \mathbb{R}^{2 \times 3}$
- D. C is undefined.

Question 3

If AB exists, which statement must be true?

- A. BA also exists.
- B. $AB = BA$.
- C. Columns of $A =$ rows of B .
- D. A and B are square.

Question 4

If $A \in \mathbb{R}^{4 \times 2}$, which identity is dimensionally correct?

A. $I_2 A = A$

B. $I_4 A = A$

C. $A I_4 = A$

D. $I_3 A = A$

Question 5

Which formula is always correct when the products exist?

A. $(AB)^T = A^T B^T$

B. $(AB)^T = B^T A^T$

C. $(AB)^T = AB$

D. $AB = BA$

Question 6

Let

$$M = \begin{pmatrix} 1 & 0 & 4 \\ 0 & 1 & -2 \\ 0 & 0 & 1 \end{pmatrix}, \quad p = \begin{pmatrix} 4 \\ -2 \end{pmatrix}.$$

Which is a valid block representation of M ?

- A. $M = \begin{pmatrix} I_2 & p \\ 0 & 1 \end{pmatrix}$
- B. $M = \begin{pmatrix} I_2 & 0 \\ p^T & 1 \end{pmatrix}$
- C. $M = \begin{pmatrix} 1 & p^T \\ 0 & I_2 \end{pmatrix}$
- D. $M = \begin{pmatrix} I_3 & p \\ 0 & 1 \end{pmatrix}$

Quiz answers

Q	Answer	Reason
1	A	Only A and B have the same size, 2×3 .
2	A	The outer dimensions remain: 3×2 .
3	C	The inner dimensions must agree.
4	B	The identity on the left must be 4×4 .
5	B	Transposing a product reverses the factor order.
6	A	The first two columns form I_2 , and the final column contains p above 1.

Final takeaway

Matrix multiplication becomes easier when dimensions and structure are read before any arithmetic begins.